

How a complaint becomes a triage level

The active Roman Urdu cardiac triage workflow: local translation, an anatomical check, sentence embeddings and structured patient features.

MODEL IDENTITY

Two separate experiments

The desktop triage classifier and the professor baseline use different encoders and datasets. Both encoders produce 384 numbers, but those vectors are not interchangeable.

GUI AND BATCH CLI

Embedding classifier, 2,252 rows

Cardiac triage using English complaint embeddings, vitals and categorical fields.

bundle	trriage_model_embedding_english/
method	C: preprocessed embeddings
encoder	MiniLM-L12-v2
classifier	Logistic Regression
inputs	410 features
output	Triage level 1-4

PROFESSOR BASELINE

Text-only classifier, 185 rows

A separate reproduction that predicts both triage and department without structured patient features.

bundle	models_src/
encoder	multilingual-e5-small
prefix	passage:
classifiers	RandomForest, two targets
inputs	384 embedding features
entry point	run_inference.py

From Roman Urdu to a classifier input

1. **Collect the complaint and structured inputs.** The GUI accepts Roman Urdu text, age, vital signs, gender, arrival mode, AVPU and ECG status. Substituted missing or unreadable structured values are disclosed.
2. **Normalize spelling.** The fuzzy clinical dictionary corrects known Roman Urdu variants before translation. It also provides inspectable text stages.
3. **Translate locally.** Ollama runs on localhost. The preferred model is llama3.2; the application reports the installed model actually selected. Requests use temperature 0.0 and few-shot examples.
4. **Apply translation guardrails.** Refusal filtering removes unusable answers. A deterministic anatomical check requires named body parts to survive translation. Failed translations or anatomical mismatches receive no triage score.
5. **Preprocess English.** Remove the active bundle's learned stop words. The English model has skip_normalization=true, so English text is not converted back through the Roman Urdu dictionary.
6. **Embed and combine features.** MiniLM produces a 384-dimensional L2-normalized vector. Concatenate it after the 26 structured features.
7. **Classify and display.** Logistic Regression produces class probabilities and triage level 1-4. The existing low-signal confidence cap and warnings remain part of scoring.

FEATURE ASSEMBLY

410 classifier inputs

NUMERIC: 6 age and vitals StandardScaler	CATEGORICAL: 20 four fields one-hot encoding	EMBEDDING: 384 MiniLM-L12-v2 L2-normalized
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The six numeric inputs are age, heart rate, systolic and diastolic blood pressure, temperature and SpO2. Gender, arrival mode, AVPU and ECG status supply the categorical columns. The embedding block is not standardized again.

The manifest must declare structured features followed by embeddings. Model loading rejects unsupported layouts and checks the classifier input width. A missing bundle does not select a replacement classifier.

EVALUATION

What the recorded scores measure

The active model was trained on 9 September 2026 using 2,252 synthetic cardiac records. The recorded stratified split contains 1,801 training rows and 451 test rows, seed 42.

CONFIGURATION	ACCURACY	UNDER - TRIAGE	OVER - TRIAGE
B: raw English embeddings	80.71%	13.30%	5.99%
C: preprocessed English embeddings	80.49%	12.64%	6.87%

These are the existing recorded results. The project cleanup retained the C classifier weights and did not rerun this experiment. Preprocessing reduced recorded under-triage by 0.66 points and accuracy by 0.22 points on this split.

Training translations were prepared using gpt-4o-mini; serving translates with Ollama. The classifier scores therefore do not measure the complete live translation workflow. Historical training also fits structured preprocessing before splitting and exports a full-data stop-word list beside a split-trained classifier. Those limitations belong in the research report.

OPERATOR MEANING

Confidence and safety checks

The displayed confidence is the winning class's probability, subject to the existing low-signal cap. It is not clinically calibrated. Anatomical consistency checks body-part preservation; it does not prove that every clinical detail or urgency cue survived translation.

The desktop exposes six tabs: Triage a Patient, Pipeline Explorer, Stop Words, Batch File, Results and Cluster Analysis. Batch output includes translation and gate verdicts. Rejected rows retain an explanation and no prediction.

The data is synthetic and the classifier is cardiac-only. The system is a research prototype, not a clinically validated medical device. Separate baseline and similarity studies cannot establish clinical reliability of its triage outputs.

Configuration sources: the active bundle's model_manifest.json and triage_metrics.json; triage_pipeline.py; src/offline_pipeline.py; train_embedding_pipeline.py.